USB e alimentazione

File: usb_alimentazione.kicad_sch

MCU

File: mcu.kicad_sch

Display (modulo 7 pin)

File: display.kicad_sch

Periferiche I2C e pulsante

File: periferiche.kicad_sch

Blocchi (vedi docs/02-architettura-hardware.md e docs/12-selezione-componenti.md):
- USB e alimentazione: USB-C 2.0 (CC 5,1 k), PTC 500 mA, ESD USBLC6-2SC6, LDO TLV75533 (stabile con MLCC), punti di test.
- MCU: ESP32-C3-MINI-1-H4X, RC su EN, disaccoppiamento; nessun tasto (DEC-039): EN e IO9 sui punti di test TP6/TP11.
- Display: header 7 pin 2,54 mm verso il modulo GC9A01 (VCC, GND, SCL, SDA, DC, CS, RST), alimentazione del modulo commutata da Q1 per spegnerlo di notte (DEC-034). Niente touch.
- Periferiche: pull-up I2C, connettori JST-SH 4 poli tipo Qwiic per SHT40 (J3) e modulo RTC DS3231 (J4), pulsante utente remoto su connettore (J5).

Assegnazione GPIO (ESP32-C3-MINI-1):
IO0, IO1 liberi su punti di test | IO2 I2C_SDA | IO3 BTN_USER | IO4 LCD_DC | IO5 LCD_RST | IO6 LCD_SCLK
IO7 LCD_MOSI | IO8 I2C_SCL | IO9 BOOT | IO10 LCD_CS | IO18 USB_D- | IO19 USB_D+ | IO20 (RXD0) DISP_EN_N
IO21 (TXD0) UART_TX su punto di test. Strapping: IO2 e IO8 con pull-up 4,7 k (I2C), IO9 pull-up 10 k.

Indirizzi I2C: SHT40 0x44, DS3231 0x68 (EEPROM assente sul modulo Adafruit).

RTC esterno: modulo Adafruit DS3231 STEMMA QT su J4 (DEC-024).
ESP32-C3-MINI-1-H4X, USB-C, LDO TLV75533, display GC9A01 su modulo 7 pin (DEC-034).
Schema gerarchico. Rev 0.2 NON validata: revisione visiva del proprietario richiesta.

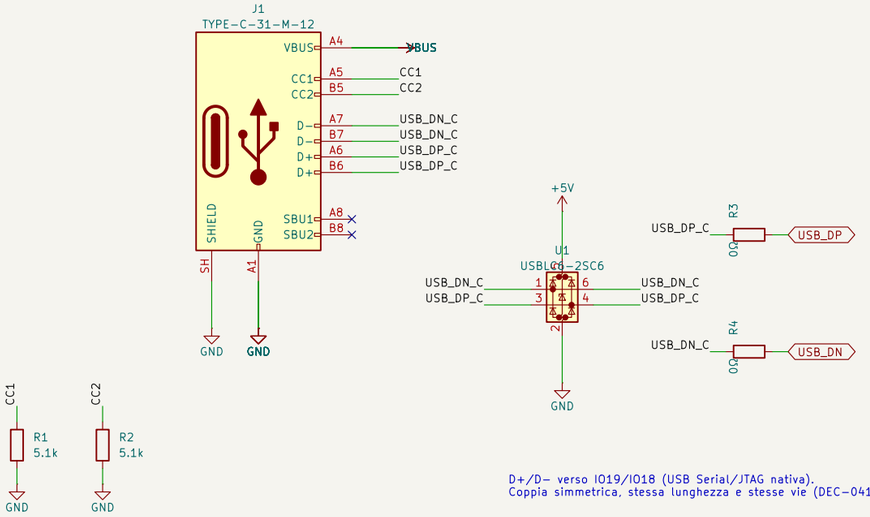
PoliClock

Sheet: /
File: policlock-main.kicad_sch

Title: PoliClock — scheda principale

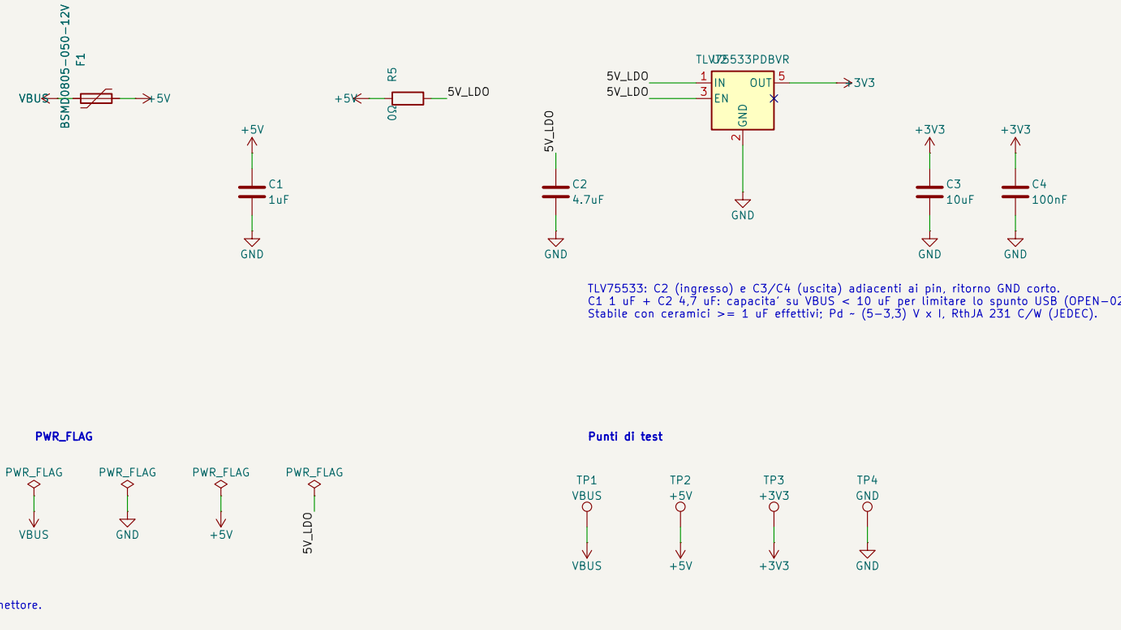
Size: A3	Date: 2026-11-08	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/5

Ingresso USB-C, CC, ESD



D+/D- verso IO19/IO18 (USB Serial/JTAG nativa).
Coppia simmetrica, stessa lunghezza e stesse vie (DEC-041), USBLC6 vicino al connettore.

Protezione VBUS e regolatore 3,3 V

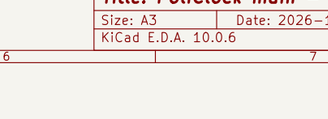


TLV75533: C2 (ingresso) e C3/C4 (uscita) adiacenti ai pin, ritorno GND corto.
C1 1 uF + C2 4,7 uF: capacita' su VBUS < 10 uF per limitare lo spunto USB (OPEN-029).
Stabile con ceramici >= 1 uF effettivi; Pd ~ (5-3,3) V x I, RthJA 231 C/W (JEDEC).

PWR_FLAG

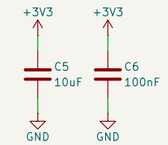


Punti di test

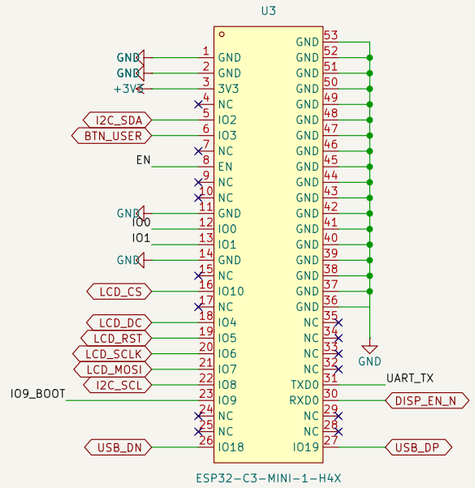


PoliClock		
Sheet: /USB e alimentazione/		
File: usb_alimentazione.kicad_sch		
Title: PoliClock main — USB e alimentazione		
Size: A3	Date: 2026-11-08	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 2/5

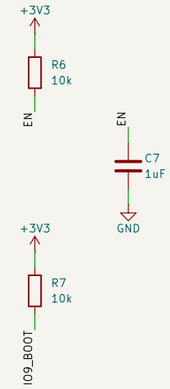
Modulo ESP32-C3-MINI-1-H4X (chip v1.1, 4 MB flash, antenna PCB oltre il bordo)



Disaccoppiamento vicino al pin 3V3 del modulo

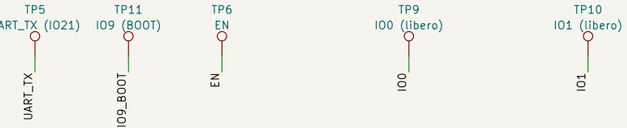


EN (reset) e IO9 (boot): senza tasti (DEC-039)



Nessun tasto (DEC-039): flash e reset via USB nativa.
Recupero: collegare TP11 (IO9) a GND e dare
alimentazione o portare TP6 (EN) a GND e rilasciarlo
-> download (IO9=0, IO8=1).

Debug



Log e flash principali via USB nativa;
TXD0 su punto di test per debug di riserva.

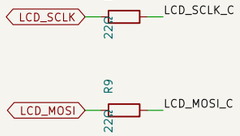
PoliClock

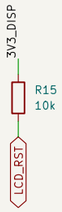
Sheet: /MCU/
File: mcu.kicad_sch

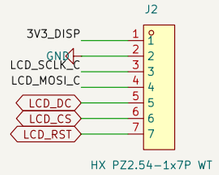
Title: PoliClock main — MCU

Size: A3	Date: 2026-11-08	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 3/5

Modulo display GC9A01 1,28" a 7 pin (DEC-034): header 2,54 mm, alimentazione commutata

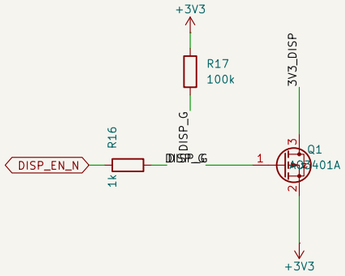


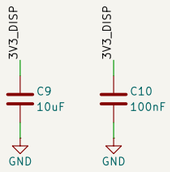




J2 -> modulo (serigrafia del modulo): 1 VCC, 2 GND, 3 SCL, 4 SDA, 5 DC, 6 CS, 7 RST.
Ordine da verificare sul modulo ricevuto prima del collegamento.
Il modulo non ha pin BLK: la retroilluminazione segue l'alimentazione VCC.

Interruttore di alimentazione del modulo (notte = spento, DEC-009):
IO20 basso -> Q1 acceso. R17 tiene Q1 spento durante reset e boot.
Con il modulo spento il firmware porta SCLK/MOSI/DC/CS/RST a livello basso.





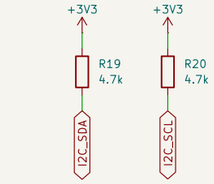
PoliClock

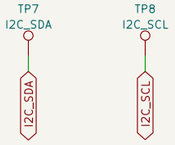
Sheet: /Display (modulo 7 pin)/
File: display.kicad_sch

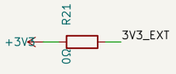
Title: PoliClock main – Display (modulo 7 pin)

Size: A3	Date: 2026-11-08	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 4/5

Bus I2C (SHT40 0x44, RTC DS3231 0x68)

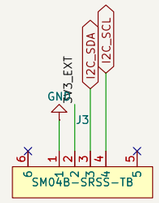


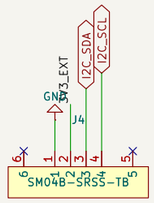




Il modulo Adafruit DS3231 ha gia' pull-up 10 k verso Vin:
in parallelo a R19/R20 -> ~3,2 k (~2,4 k se anche il breakout SHT40
ha pull-up 10 k, da verificare), accettabile.

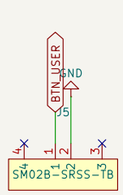
Connettori Qwiic / STEMMA QT (JST-SH 4 poli: 1 GND, 2 3V3, 3 SDA, 4 SCL)

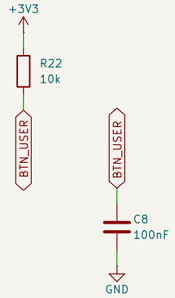




J3: modulo breakout SHT40 commerciale (header 2,54 mm), cavo JST-SH 4 poli -> jumper
femmina; ordine dei pin dalla serigrafia del modulo (DEC-028).
J4: modulo Adafruit DS3231 Precision RTC STEMMA QT (5188) con pila CR1220 (DEC-024).

Pulsante utente





IO3: risveglio da deep-sleep possibile (GPIO0-5).
Solo pulsante remoto su J5 (DEC-032): nessun tasto USER su scheda.

PoliClock		
Sheet: /Periferiche I2C e pulsante/		
File: periferiche.kicad_sch		
Title: PoliClock main — Periferiche I2C e pulsante		
Size: A3	Date: 2026-11-08	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 5/5